Group: Adios

Domain 2

PRODUCT REQUIREMENT DOCUMENT (PRD)

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Project Name: Ka-ching AI

1. Project Overview

- **Problem Statement:**
  Many businesses struggle to make optimal decisions on resource allocation, product mix, and budgeting due to fragmented data and reliance on intuition. This often leads to inefficiencies, waste, and missed profit opportunities. Existing solutions are either too complex or not flexible across industries. Therefore, there is a need for a scalable, AI-driven decision support system that can analyze business data, consider constraints, and provide actionable insights to improve efficiency and maximize profitability across any sector.
- **Target Domain:**
  SMEs and growing businesses across diverse industries that lack access to advanced analytics tools and require an affordable, AI-driven decision intelligence system to optimize operations, improve profitability, and support smarter business decisions.
- **Proposed Solution Summary:**
  This project proposes an **AI-driven decision intelligence system** designed to help businesses optimize their operations through data-driven insights. The system integrates key business inputs—such as resources, inventory, budget, product performance, and operational constraints—and analyzes them using intelligent algorithms.
  Based on this analysis, the system generates **actionable recommendations** including optimal resource allocation, product mix adjustments, budget redistribution, and opportunity cost insights. It also provides **profit forecasting and waste detection**, enabling businesses to evaluate the impact of different strategies before implementation.
  The solution is **industry-agnostic and scalable**, making it suitable for SMEs across various sectors. By transforming complex data into clear, explainable decisions, the system empowers businesses to **maximize profit, improve efficiency, and reduce waste** without requiring advanced technical expertise.

2. Background & Business Objective

- **Background of the Problem:**

  In today's competitive and fast-changing business environment, organizations are required to make frequent operational and financial decisions involving resource allocation, inventory management, product prioritization, and budgeting. However, many businesses—especially SMEs—still rely on manual processes, spreadsheets, or intuition to guide these decisions. While large amounts of business data are available, they are often **fragmented across different functions**. This situation leads to common issues such as **inefficient use of labor and machinery, excess or insufficient inventory, poor product mix decisions, and underutilization of budgets**. Additionally, most businesses are unable to assess the **opportunity cost** of their decisions or accurately forecast the impact on profitability.

  Although advanced analytics solutions exist, they are often **costly, complex, or industry-specific**, making them inaccessible or unsuitable for many SMEs. This creates a gap between data availability and actionable insights, highlighting the need for a more accessible, intelligent, and flexible decision-support system.

- **Importance of Solving This Issue:**

  For SMEs, the impact is even more significant due to limited resources and tighter margins. Without proper decision support, small inefficiencies can accumulate into substantial financial losses, making it difficult for businesses to compete and scale.

  Solving this issue enables businesses to:

  - Make **data-driven decisions** instead of relying on intuition
  - **Maximize profit** through better product and resource optimization
  - **Reduce waste and operational inefficiencies**
  - Quickly adapt to changing market demand and constraints
  - Improve overall productivity and competitiveness

- **Strategic Fit / Impact:**

  The solution delivers significant **economic and operational benefits**, especially for SMEs:

  - **Increased profitability** through optimized decisions
  - **Reduced waste** in inventory and resources
  - **Improved efficiency** across operations
  - **Better accessibility to AI tools** for businesses with limited expertise
  - **Economic empowerment** by enabling smarter, faster decision-making

3. Product Purpose

3.1. Main Goal of the System:

The main goal of the system is to develop an **AI-powered decision intelligence platform** that helps businesses optimize their operations by transforming raw business data into **clear, actionable, and data-driven recommendations**.

The system aims to support organizations in making better decisions on **resource allocation, product mix, budgeting, and operational planning**, while considering real-world constraints such as capacity, inventory, and demand.

Ultimately, the system seeks to **maximize profitability, improve efficiency, reduce waste, and enhance decision-making quality** across different industries in an accessible and scalable way.

3.2. Intended Users (Target Audience):

- Business owners and entrepreneurs who manage daily operations and strategic decisions
- Operations managers responsible for resource allocation, production and efficiency
- Finance and budgeting teams
- Marketing teams
- Inventory and supply chain managers

4. System Functionalities

4.1. Description:

The system is an AI-powered decision intelligence platform that uses a **Prompt-to-UI model** to convert business inputs into interactive dashboards and actionable insights. It processes company data (resources, products, budget, and constraints) and automatically analyzes relationships to generate **optimized recommendations** such as resource reallocation, product mix changes, and cost improvements. The system also provides **profit forecasting, opportunity cost analysis, and waste detection**, while dynamically generating multi-screen UI views to present insights clearly and intuitively.

4.2. Key Functionalities:

- **Business Decision Setup**
  Transforms high-level business objectives (e.g., maximize profit, optimize capacity) into structured analysis workflows by allowing users to define sector, goals, constraints, and time horizon upfront.

- **Opportunity Cost & Option Comparison**
  Automatically evaluates and ranks multiple business or promotional options using key metrics such as profit margin, ROI, and risk score, while quantifying the opportunity cost of selecting suboptimal choices.

- **AI-Powered Recommendation Engine**
  Generates clear, actionable, and explainable recommendations using LLM-based intelligence (e.g., ILMU GLM-5.1), including confidence scores and detailed reasoning behind each decision.

- **What-If Scenario Simulation**
  Enables users to test changes in variables such as demand, cost, or capacity, and instantly observe how optimal decisions and outcomes shift under different scenarios.

- **Decision History Tracking**
  Maintains a structured history of all analyses and recommendations, allowing users to review past decisions, compare outcomes, and improve future strategies.

- **Sector-Agnostic Data Input**
  Supports flexible input for multiple business options across any industry, with customizable fields and optional demo data auto-fill to accelerate system usage and testing.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

Ilmu GML-5.1(Z.AI GLM family) was selected as the core LLM for OpporTune AI due to three key capabilities:

1. **Domain Adaption for Business & Operational Data**: The model is optimized for structured business data interpretation, including profit/loss metrics, margin calculations, and risk scoring, which are core to the product's F&B, manufacturing, and retail sector use cases.
2. **Multi-Sector Generalization:** Unlike narrow industry-specific models, Ilmu GLM-5.1 supports consistent performance across diverse business sectors.
3. **AI-Powered Recommendation Engine**
   Generates clear, actionable, and explainable recommendations using LLM-based intelligence (e.g., ILMU GLM-5.1), including confidence scores and detailed reasoning behind each decision.
4. **What-If Scenario Simulation**
   The model's fine-tuning for factual business analysis minimizes hallucinations in critical tasks like opportunity cost calculation and risk assessment. Enables users to test changes in variables such as demand, cost, or capacity, and instantly observe how optimal decisions and outcomes shift under different scenarios.

5. **Decision History Tracking**
   Maintains a structured history of all analyses and recommendations, allowing users to review past decisions, compare outcomes, and improve future strategies.
6. **Localization & Language Support**: As a Southeast Asian-focused model, it natively handles regional currencies (e.g., MYR/RM) and market terminology, ensuring recommendations are culturally and financially relevant to the product's target user base.

4.3.2. Prompting Strategy:

Multi-step agentic prompting (core strategy):
The system breaks down a user's request into sequential reasoning steps (data interpretation → constraint analysis → option comparison → optimization → recommendation generation). Each step is handled by a specialized "agent-like" prompt stage. Decisions involve multiple stages such as resource allocation, constraint checking, opportunity cost analysis, and profit forecasting. A single prompt would not reliably handle this complexity.

Few-shot prompting (support layer):
We provide example inputs and expected outputs for business scenarios (e.g., F&B, manufacturing, cosmetics) to guide the model in producing consistent, structured, and domain-agnostic responses. This ensures the model produces structured outputs across different industries and input formats.

Agentic flow improves accuracy and interpretability, making the system behave like a decision engine rather than a chatbot, which is critical for business users who need explainable insights.

Chain-of-Thought Prompting:

Ideal for the system in decision-making: analyze constraints, compare options, calculate opportunity cost, generate recommendation etc. This improves transparency and accuracy of business decisions.

Tree-of-Thought Prompting:

Explore multiple decision paths before selecting the best one. This is very suitable for use case because businesses often have multiple competing strategies.

Retrieval-Augmented Prompting:

Use external structured data to guide responses based prompting ensures decisions are grounded in real business data instead of assumptions.

4.3.3. Context & Input Handling:

Our system, **Ka-ching AI**, is built with a **decision intelligence layer on top of imperfect and unstructured SME data**. To ensure robustness, we implement a **3-level input strategy** that balances flexibility, accuracy, and performance.

1. Input Strategy Design

Level 1: Basic Mode

- Manual input only
- Supports up to **5 decision options**
- ~8 mandatory fields per option
- No truncation required due to controlled structure

Level 2: Advanced Mode

- Manual input with richer business context
- Supports up to **5 decision options**
- ~25–30 fields per option
- Includes **business notes**

Users can input unstructured data such as:

- Market conditions
- Customer behavior
- Supplier issues
- Seasonal demand
- Campaign context

Level 3: Pro Data Import (Advanced / Automated Data)

- Supports:
  - CSV / Excel uploads
  - Integration (Shopee, POS, ERP — future-ready)
  - AI extraction from natural language inputs

Limits:

- Maximum **1,000 rows per upload**
- Maximum **5 MB file size**
- Only **key columns are extracted**

2. Oversized Input Handling

- Prompts user to select top 5 most relevant options
- Notes exceed 500 characters, system automatically summarizes and keeps only decision-critical information
- CSV exceeds limits of 1000 row or 5 MB, user is prompted to reduce dataset size
- Excessive optional data: system filters and keeps only decision-relevant field such as price, cost, demand, capacity, risk and summarized notes.

3. Unstructured Inputs

Users can enter business notes such as market condition, customer behaviour, supplier issues, seasonal demand and campaign context. These notes are used by Ilmu GLM-5.1 to understand business context and explain the decisions.

4.3.4. Fallback & Failure Behavior:

1. **Automated Validation Layer**

Before presenting any AI-generated output to the user, the system performs a structured validation process to detect inconsistencies or hallucinations.

- **Constraint Validation:** Ensures recommendations do not exceed business constraints (e.g., machine hours, budget, capacity).
- **Logical Consistency Check:** Verifies financial feasibility (e.g., profit cannot exceed unrealistic bounds given inputs).
- **Schema Validation:** Ensures outputs follow required structured formats (e.g., JSON fields for profit, ROI, recommendations).

If the output is detected as:

- Off-topic
- Incomplete
- Logically inconsistent
- Likely hallucinated

Any output failing validation is flagged as invalid and not directly shown to the user.

2. **Human-in-the-Loop Escalation**

For enterprise or critical use cases:

- Users can flag questionable outputs via a “Report Issue” feature.
- The case is logged for review in an admin/monitoring dashboard.
- Feedback can be used for model improvement and future fine-tuning.

3. **Automatic Retry Mechanism**

When a response fails validation, the system automatically triggers a re-prompt with refined instructions. The retry prompt includes original user input or detected error type (e.g., missing constraint usage).

4. Structured Fallback Response

If retries fail, the system switches to a safe fallback state:

- Provides partial insights only from verified input data
- Avoids making speculative or hallucinated recommendations

5. User Stories & Use Cases

- **User Stories:**

1. SME Owner Making Product Decision

As an SME owner running a small online business, I often struggle to decide which product to sell next because I have limited capital and cannot afford to make wrong decisions. I want to input a few product options along with their estimated cost, selling price, and demand so that the system can analyze and recommend the most profitable option. This helps me reduce financial risk and make confident decisions without relying on guesswork.

2. Beginner Entrepreneur with Limited Knowledge

As a beginner entrepreneur who does not have much experience or structured data, I want to describe my situation in simple, natural language, such as "I have a low budget but want to sell trending products," so that the system can still understand my intent and provide useful recommendations. This allows me to start a business without needing technical knowledge or advanced data analysis skills.

3. Data-Driven Seller Using Real Data

As a more advanced seller who already has sales data, I want to upload my CSV or Excel files containing product performance, costs, and demand trends so that the system can generate more accurate and data-driven insights. This enables me to move beyond assumptions and base my decisions on actual business performance.

4. Risk-Aware Business Owner

As a business owner who is concerned about potential risks, I want to understand not only which option is the most profitable but also the risks and trade-offs involved, such as supplier instability or fluctuating demand. This helps me make balanced decisions that consider both short-term gains and long-term sustainability

5. Time-Constrained User

As a busy SME owner who needs to make quick decisions, I want the system to provide fast and clear recommendations without requiring me to manually analyze large amounts of data. This allows me to save time and focus on running my business while still making informed decisions.

6. Non-Technical User

As a user with little to no technical background, I want a simple, step-by-step interface that guides me through the input process so that I can use the system بسهولة without confusion or training. This ensures that I can benefit from AI decision support even without prior experience.

7. User Handling Real-World Uncertainty

As a user facing real-world uncertainties such as changing market trends or unreliable suppliers, I want to include these factors as notes in my input so that the system can take them into account when generating recommendations. This ensures that the output reflects actual business conditions rather than ideal assumptions.

8. User Comparing Multiple Opportunities

As a user evaluating multiple business opportunities, I want to compare up to five options at once and understand not only which one is the best but also what I might lose by not choosing the others. This helps me clearly see the opportunity cost and make more strategic decisions.

These user stories demonstrate how Ka-ching AI supports different types of users from beginners to experienced sellers by transforming their real-world challenges into clear, actionable business decisions.

- **Use Cases (Main Interactions):**

Prompt to UI Flow:

Users begin by entering their business problem in their own way, either through structured fields or natural language input. They can provide decision options along with key data such as cost, price, and demand, and optionally include business notes like market trends or supplier issues.

The system then parses the input to identify key variables and intent. It performs data cleaning, validation, and summarization of unstructured notes, converting them into decision-relevant signals. After preprocessing, the backend calculates important metrics such as profit and opportunity cost.

Only the essential processed data is sent to the AI model (Ilmu GLM-5.1), which evaluates the options and generates a recommendation along with risk analysis and trade-offs.

The system transforms raw user input into a clear, structured, and explainable business decision, without requiring manual analysis.

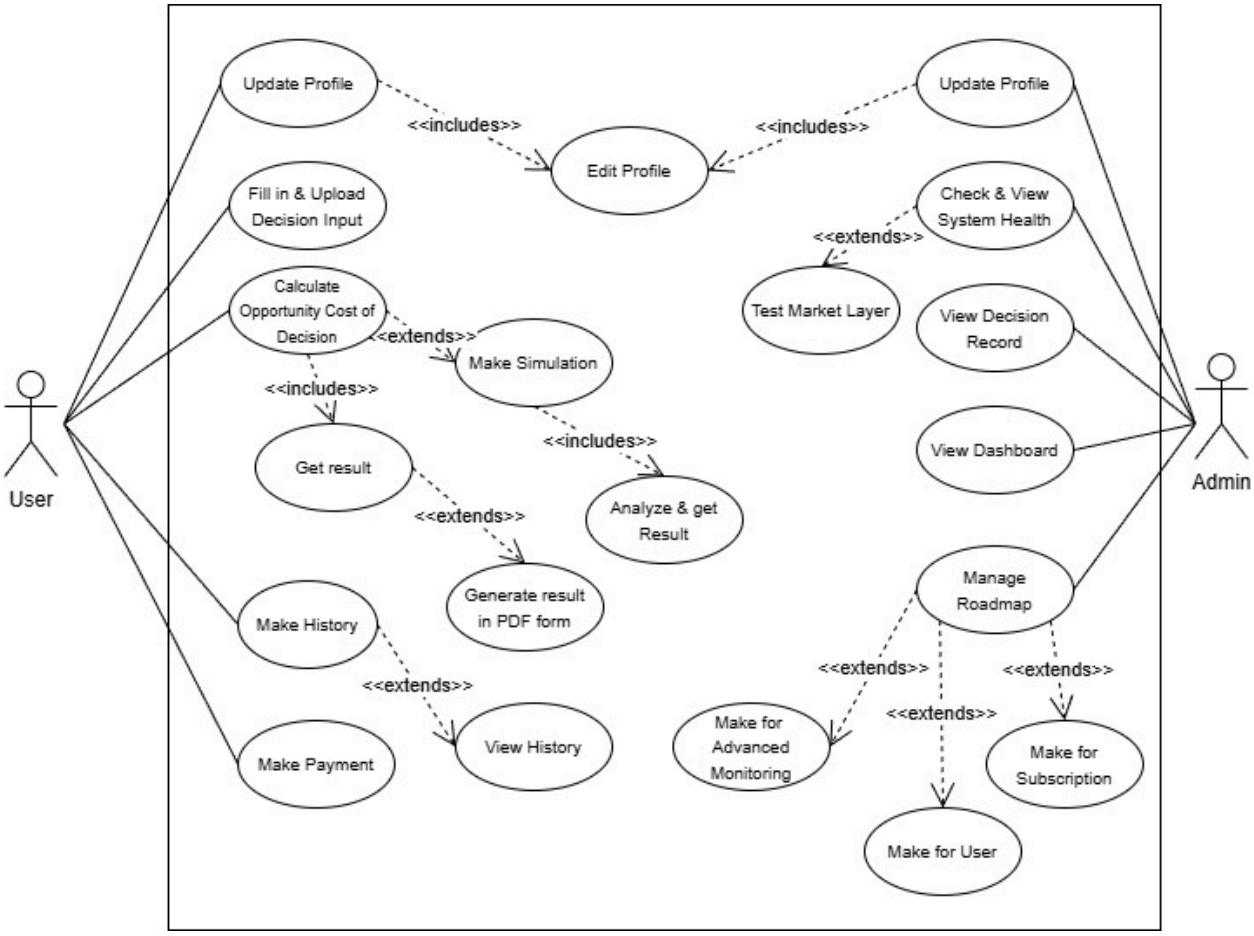
Flow Generation:

After receiving the initial recommendation, users can refine or explore further decisions by adjusting inputs, adding more context or modifying options. For example, users may update pricing or cost values, add new business notes or replace or reorder decision options.

The system reprocesses the updated inputs while maintaining consistency in evaluation logic. It then generates an updated recommendation that reflects the new scenario, ensuring continuity in reasoning and comparison.

Users can iteratively improve their decisions with the system maintaining consistent logic and context-aware insights, enabling more accurate and confident business choices.

Use Case Diagram:



6. Features Included (Scope Definition)

1. **Multi-Level Input System**
   - **Level 1 (Basic Mode):** Quick manual input for fast decision-making
   - **Level 2 (Advanced Mode):** Rich input with optional business context and notes
   - **Level 3 (Pro Mode):** CSV/Excel upload and future integration with Shopee / POS / ERP
2. **Decision Intelligence Engine**

   AI-powered evaluation using Ilmu GLM-5.1 generates opportunity analysis, risk assessment, trade-offs and recommended actions. The system analyzes all inputs including structured and unstructured. These are not just predictions, these data are explainable business decisions according to the real-world dataset and cases.

3. **Opportunity Cost Calculator**

   The system compares up to **5 decision options** and calculates the **opportunity cost**. It shows what the user gains vs what they give up and highlights the second-best alternative.

4. **Unstructured Context Understanding**

   Users can input free-text business notes such as market trends, customer behaviour and supplier issues. Then, the AI extracts meaning and incorporates it into decisions.

5. **Smart Input Validation & Optimization**

   The system ensures data quality before analysis by cleaning messy inputs, summarizing long notes and filtering irrelevant data.

6. **Structured Decision Output Dashboard**

   Results are presented in clear, user-friendly format with best option highlighted, alternatives ranked and explanation included. Users don't need to interpret raw data and they can get ready-to-use decisions.

7. **Lightweight & Scalable System Design**

   The backend sends only essential data to the AI model which reduces processing cost, improves speed and minimizes hallucination risk. Even with large datasets, users still get fast and stable responses.

8. **Future-Ready Integrations**

   Designed to connect with e-commerce platforms (Shopee, TikTok Shop), POS systems and ERP systems. In the future, users won't need manual input and the system can auto-fetch real business data.

9. **Fail-Safe & User Guidance System**

   The system handles invalid or excessive inputs by rejecting oversized data, suggesting corrections and guiding users step-by-step.

10. **SME-Centric UX Design**

    The interface is designed for non-technical users such as simple step-by-step workflow, minimal jargon and focus on actionable insights. Even first time users can go from input to decision in minutes without training. This also increases real-world adoption and impact and reduces dependency on analysis or experts.

Ka-Ching AI transforms messy SME data into clear, explainable business decisions through a structured input system, AI-powered opportunity cost analysis, and intelligent data handling.

7. Features Not Included (Scope Control)

To ensure feasibility within hackathon constraints, the current version of **Ka-ching AI** focuses on core decision intelligence features while intentionally excluding the following:

1. Full Production Backend & Infrastructure

Not included:

- Scalable backend architecture
- Secure API management
- Production-grade database systems

Outcome:

- Faster development and prototyping during hackathon
- Core AI logic is demonstrated without infrastructure overhead
- Ready to be extended into production in future phases

2. Real-Time External System Integration

Not included :

- Live integration with Shopee, TikTok Shop
- POS / ERP system connections

Outcome:

- Avoids dependency on external APIs during demo
- Ensures system stability and consistency
- Integration remains **future-ready but not required for MVP**

3. Multi-User Collaboration

Not included:

- Multiple users interacting with the same workspace simultaneously
- Team collaboration features

Outcome:

- Simplifies system design and reduces synchronization complexity
- Focus remains on **single-user decision accuracy**
- Can be scaled to multi-user in future versions

4. Large-Scale Data Processing

Not included:

- Handling datasets beyond:
    - 1,000 rows
    - 5 MB file size

Outcome:

- Maintains fast and responsive performance
- Prevents system overload during demo
- Keeps focus on **SME-scale practical usage**

5. Fully Automated Decision Execution

Not included:

- Auto-executing decisions (e.g., auto-purchasing stock, auto-listing products)

Outcome:

- System remains a **decision-support tool**, not a control system
- Keeps users in control of final business actions
- Reduces risk of unintended consequences

6. Advanced Predictive Modeling / Long-Term Forecasting

Not included:

- Deep predictive analytics (e.g., multi-year forecasting, complex simulations)

Outcome:

- Focus on **short-term, actionable decisions**
- Faster response and easier interpretation for SMEs
- Avoids over-complexity and unreliable long-term predictions

8. Assumptions & Constraints

1. LLM Cost Constraint

Estimated Token Usage per Session:

- Average session: ~2,000 – 4,000 tokens
    - Input (user + processed data): ~1,000–1,500 tokens
    - Output (analysis + explanation): ~1,000–2,500 tokens

Estimated Cost Consideration: Designed to remain low-cost per SME user session and scalable for high usage

Cost Optimization Design Decisions:

- Data Filtering Before AI Call
  Only essential fields are sent: price, cost, demand, risk, summarized notes
  - Reduces unnecessary token usage
- Summarization of Unstructured Inputs
  Long notes are compressed into short decision-relevant context to prevents token explosion
- Option Limitation (Max 5 options)
  Restricts comparison scope to keep prompt size predictable and efficient
- No Raw CSV Sent to LLM
  Only aggregated insights are passed and major reduction in token consumption
- (Optional Future) Caching Repeated Queries
  Similar decision scenarios can reuse previous outputs for further reduced cost at scale

Outcome:

- Predictable and controlled inference cost
- Scalable for SMEs without high operational expenses

2. Technical Constraints

Current Limitations:

- No full backend production infrastructure
- No real-time API integrations (Shopee, POS, ERP) in MVP
- CSV upload limited to 1,000 rows / 5 MB
- No multi-user collaboration support

  Currently, the system operates on **static CSV/JSON uploads** or mock data. Real-time bank synchronization is out of scope for this MVP to ensure data privacy and avoid the complexity of financial regulatory sandboxes.

Impact:

- System operates as a functional prototype (MVP)
- Focus remains on validating decision intelligence logic, not full deployment

3. Performance Constraints

- AI response depends on:
  - Number of decision options
  - Amount of contextual input

Constraints applied:

- Maximum 5 decision options
- Notes limited to 500 characters per option
- Large inputs are summarized or rejected

Impact:

- Ensures fast response time (near real-time)
- Avoids latency caused by oversized prompts
- Maintains consistent performance during demo

4. User Input Constraints

- System requires user-provided input
- AI does not autonomously generate decisions without context

Additional constraints:

- Users must define options and provide basic business data
- Final decisions require human validation and approval

Impact:

- Keeps user in control of business decisions
- Prevents over-reliance on AI
- Ensures practical and responsible usage

Ka-ching AI is designed with strict cost, performance, and input constraints to ensure it remains scalable, efficient, and practical for SMEs, while avoiding the common pitfalls of high-cost and unreliable AI systems.

9. Risks & Questions Throughout Development

During the development of **Ka-ching AI**, several risks and design concerns were identified to ensure the system remains reliable, scalable, and suitable for real-world SME decision-making. These risks mainly relate to AI consistency, data quality, interpretation of unstructured inputs, and user behavior. The following points summarize these concerns in a structured form, while still maintaining an overall system perspective.

1. Decision Similarity Risk

A key concern is that the AI may generate **similar recommendations for different users** even when their business contexts are different. This reduces personalization and trust in the system. To mitigate this, Ka-ching AI integrates **strong contextual inputs** such as budget, cost structure, risk preference, and unstructured business notes. These factors ensure that each recommendation is tailored to the **user's unique situation**, while opportunity cost analysis further differentiates outputs based on **real trade-offs**.

2. Decision Stability and Consistency

Another risk is inconsistency in AI outputs when similar inputs are provided. This can negatively affect user confidence. To address this, the system standardizes inputs through preprocessing and **limits decision complexity** by restricting the number of options and enforcing structured fields. In addition, **deterministic calculations** such as profit estimation and opportunity cost are combined with **AI explanations** to ensure that outputs remain stable and repeatable.

3. Misinterpretation of Unstructured Inputs

Users may provide business notes in natural language, which introduces the risk of misinterpretation by the AI. This could lead to incorrect insights if not properly managed. To reduce this risk, all unstructured inputs are **first cleaned, summarized, and converted into key decision-relevant signals** before being processed by the model. This ensures that only meaningful and concise context is used in generating outputs.

4. Data Quality Risk

Since SME **users** may input **incomplete or estimated data**, poor input quality could directly affect output accuracy. To mitigate this, the system **enforces validation rules** and ensures that **essential fields** such as cost, price, and demand are always provided. The system also guides users when data is missing or inconsistent, improving overall input reliability.

5. Over-Reliance on AI Decisions

There is a risk that users may blindly follow AI recommendations without critical thinking. To prevent this, Ka-ching AI is designed strictly as a decision-support system rather than an automated decision-maker. Every output includes **explanations, trade-offs, and alternative options** so that users can understand the reasoning and retain full control over final decisions.

6. Scalability and Performance Constraints

Handling large datasets or complex inputs may impact system performance and increase computational cost. To manage this, strict constraints are applied, such as **limiting decision options and file sizes**. Additionally, only essential processed data is sent to the AI model, ensuring efficient processing and maintaining **fast response** times even under higher usage.

7. Edge Case and Input Ambiguity Risk

In cases where user input is unclear, conflicting, or incomplete, the system may produce unreliable outputs. To handle this, Ka-ching AI includes **fail-safe mechanisms** that either reject invalid inputs or prompt users to refine their data before processing. This ensures that only meaningful and usable information is passed to the AI.

Overall, these risks are carefully addressed through **structured input design, preprocessing, validation, and controlled AI usage**. This ensures that Ka-ching AI remains reliable, explainable, and practical for SME decision-making, even when working with imperfect real-world data.